WEAPONS AND TECHNOLOGY

NAUTICAL INSTRUMENTS

BY THE 19TH CENTURY navigation at sea had reached a high level of accuracy. Previously, seamen had mostly kept within sight of shore, depending upon observation of the sun and stars for orientation. The introduction of the compass and of instruments such as the astrolabe by Chinese and Arab mariners in medieval times were major advances on which European navigators systematically built.



Early astrolabe Thirteenth-century navigators use an astrolabe in the Indian Ocean

TYPES OF INSTRUMENT

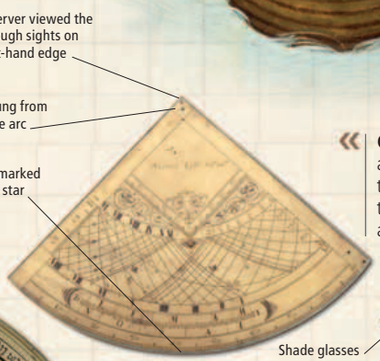
The compass was a difficult instrument to use at sea because of subtle and complex variations between readings of magnetic and true north. European mariners had made great progress in adjusting compass readings to true north by the end of the 16th century. The navigator's other prime instruments were devices such as the astrolabe, or the more advanced sextant for measuring the altitude of heavenly bodies such as the Pole Star or the Sun at noon. With the aid of astronomical almanacs this allowed him to calculate his latitude—how far his ship was north or south of the equator. It took until the mid-18th century for longitude to be calculated accurately.



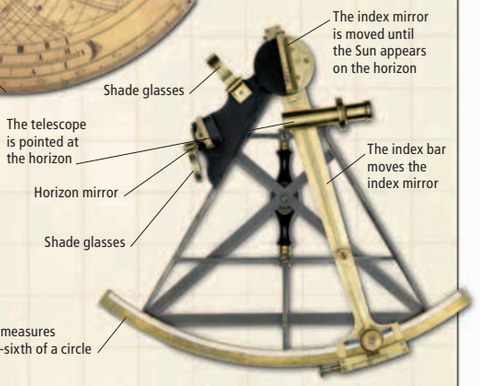
COMPASS The magnetic compass was in use in China from the 11th century. It needed skill and experience to use a compass accurately on oceanic voyages.



ASTROLABE The astrolabe was used by medieval astronomers to observe the stars. A simplified version was adopted by mariners.



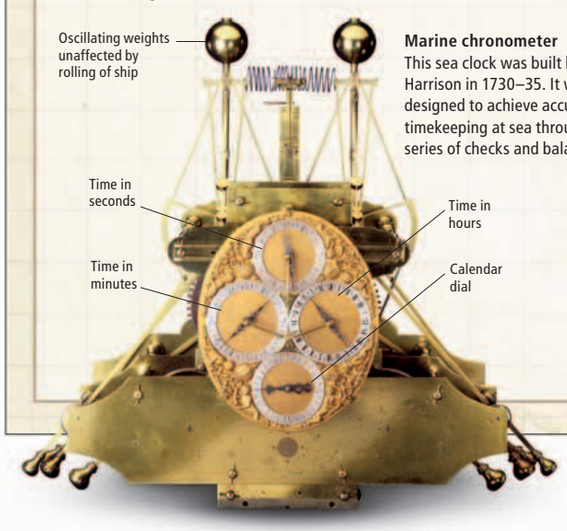
QUADRANT The quadrant was a primitive instrument for finding the altitude of stars, including the Sun. It was hard to use accurately on a rolling ship.



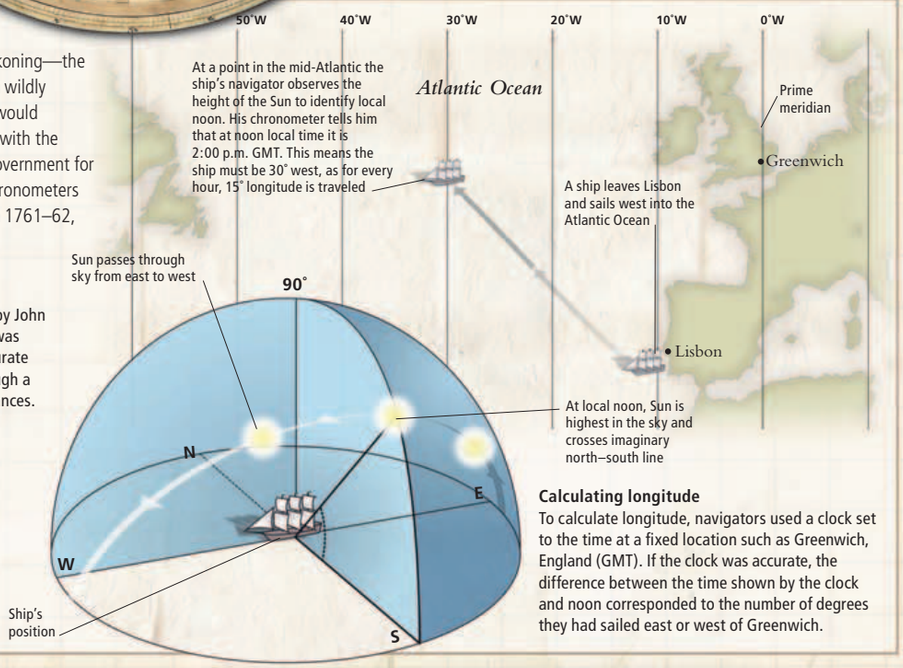
SEXTANT Invented in the mid-18th century, the sextant was a great improvement over the astrolabe for measuring the altitude of the Sun.

MEASURING LONGITUDE

Until the 18th century calculations of longitude depended on dead reckoning—the measurement of the ship's speed and direction of travel. This was often wildly inaccurate. A more precise calculation required the use of a clock that would keep accurate time at sea, but conventional timepieces could not cope with the movement of the ship. In response to a reward offered by the British government for solving this problem, clockmaker John Harrison produced a series of chronometers of increasing accuracy. His H4, tested on a voyage to the West Indies in 1761–62, allowed longitude to be calculated to within one nautical mile.



Marine chronometer This sea clock was built by John Harrison in 1730–35. It was designed to achieve accurate timekeeping at sea through a series of checks and balances.



Calculating longitude To calculate longitude, navigators used a clock set to the time at a fixed location such as Greenwich, England (GMT). If the clock was accurate, the difference between the time shown by the clock and noon corresponded to the number of degrees they had sailed east or west of Greenwich.